

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Monday 5 Oct 2020

Afternoon (Time: 1 hour 40 minutes)

Paper Reference **8FM0/01**

Further Mathematics

Advanced Subsidiary

Paper 1: Core Pure Mathematics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Given that

$$\begin{aligned} z_1 &= 2 + 3i \\ |z_1 z_2| &= 39\sqrt{2} \\ \arg(z_1 z_2) &= \frac{\pi}{4} \end{aligned}$$

where z_1 and z_2 are complex numbers,

(a) write z_1 in the form $r(\cos \theta + i \sin \theta)$

Give the exact value of r and give the value of θ in radians to 4 significant figures.

(2)

(b) Find z_7 , giving your answer in the form $a + ib$ where a and b are integers.

(6)

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



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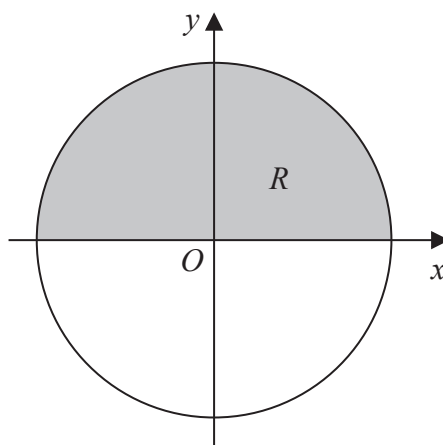


Figure 1

Figure 1 shows a circle with radius r and centre at the origin.

The region R , shown shaded in Figure 1, is bounded by the x -axis and the part of the circle for which $y > 0$

The region R is rotated through 360° about the x -axis to create a sphere with volume V

Use integration to show that $V = \frac{4}{3}\pi r^3$

(5)

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4.

All units in this question are in metres.

A lawn is modelled as a plane that contains the points $L(-2, -3, -1)$, $M(6, -2, 0)$ and $N(2, 0, 0)$, relative to a fixed origin O .

- (a) Determine a vector equation of the plane that models the lawn, giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ (3)

- (b) (i) Show that, according to the model, the lawn is perpendicular to the vector $\begin{pmatrix} 1 \\ 2 \\ -10 \end{pmatrix}$
- (ii) Hence determine a Cartesian equation of the plane that models the lawn.
- (4)**

There are two posts set in the lawn.

There is a washing line between the two posts.

The washing line is modelled as a straight line through points at the top of each post with coordinates $P(-10, 8, 2)$ and $Q(6, 4, 3)$.

- (c) Determine a vector equation of the line that models the washing line. **(2)**
- (d) State a limitation of one of the models. **(1)**

The point $R(2, 5, 2.75)$ lies on the washing line.

- (e) Determine, according to the model, the shortest distance from the point R to the lawn, giving your answer to the nearest cm. (2)

Given that the shortest distance from the point R to the lawn is actually 1.5 m,

- (f) use your answer to part (e) to evaluate the model, explaining your reasoning. (1)



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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



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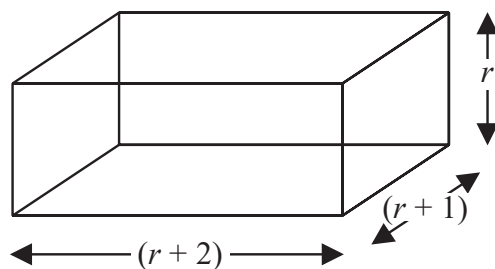


Figure 2

A block has length $(r + 2)$ cm, width $(r + 1)$ cm and height r cm, as shown in Figure 2.

In a set of n such blocks, the first block has a height of 1 cm, the second block has a height of 2 cm, the third block has a height of 3 cm and so on.

- (a) Use the standard results for $\sum_{r=1}^n r^3$, $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that the **total** volume, V , of all n blocks in the set is given by

$$V = \frac{n}{4}(n+1)(n+2)(n+3) \quad n \geq 1 \quad (5)$$

Given that the total volume of all n blocks is

$$(n^4 + 6n^3 - 11\,710) \text{ cm}^3$$

- (b) determine how many blocks make up the set. (2)

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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6. (i)

$$\mathbf{A} = \begin{pmatrix} 2 & a \\ a-4 & b \end{pmatrix}$$

where a and b are non-zero constants.

Given that the matrix $\mathbf{A}$ is self-inverse,

- (a) determine the value of b and the possible values for a . (5)

The matrix $\mathbf{A}$ represents a linear transformation M .

Using the smaller value of a from part (a),

- (b) show that the invariant points of the linear transformation M form a line, stating the equation of this line. (3)

(ii)

$$\mathbf{P} = \begin{pmatrix} p & 2p \\ -1 & 3p \end{pmatrix}$$

where p is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation U .

The triangle T has vertices at the points with coordinates $(1, 2)$, $(3, 2)$ and $(2, 5)$.

The area of the image of T under the linear transformation U is 15

- (a) Determine the value of p . (4)

The transformation V consists of a stretch scale factor 3 parallel to the x -axis with the y -axis invariant followed by a stretch scale factor -2 parallel to the y -axis with the x -axis invariant. The transformation V is represented by the matrix $\mathbf{Q}$.

- (b) Write down the matrix $\mathbf{Q}$. (2)

Given that U followed by V is the transformation W , which is represented by the matrix $\mathbf{R}$,

- (c) find the matrix $\mathbf{R}$. (2)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 16 marks)



7.

$$f(z) = z^4 + az^3 + bz^2 + cz + d$$

where a , b , c and d are real constants.

The equation $f(z) = 0$ has complex roots z_1 , z_2 , z_3 and z_4

When plotted on an Argand diagram, the points representing z_1 , z_2 , z_3 and z_4 form the vertices of a square, with one vertex in each quadrant.

Given that $z_1 = 2 + 3i$, determine the values of a , b , c and d .

(6)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8. Prove by induction that, for $n \in \mathbb{Z}^+$

$$f(n) = 2^{n+2} + 3^{2n+1}$$

is divisible by 7

(6)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9. The cubic equation

$$3x^3 + x^2 - 4x + 1 = 0$$

has roots α , β , and γ .

Without solving the cubic equation,

(a) determine the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (3)

(b) find a cubic equation that has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$, giving your answer in the form

$x^3 + ax^2 + bx + c = 0$, where a , b and c are integers to be determined. (3)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10. Given that there are two distinct complex numbers z that satisfy

$$\{z: |z - 3 - 5i| = 2r\} \cap \left\{z: \arg(z - 2) = \frac{3\pi}{4}\right\}$$

determine the exact range of values for the real constant r .

(7)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 7 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. The cubic equation

$$2x^3 + 6x^2 - 3x + 12 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$, giving your answer in the form $pw^3 + qw^2 + rw + s = 0$, where p , q , r and s are integers to be found.

(5)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$

(6)



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Question 3 continued

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(Total for Question 3 is 6 marks)



4. The line l has equation

$$\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$$

The plane Π has equation

$$\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -7$$

Determine whether the line l intersects Π at a single point, or lies in Π , or is parallel to Π without intersecting it.

(5)



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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 5 marks)



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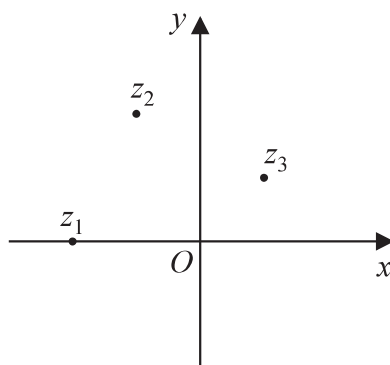


Figure 1

The complex numbers $z_1 = -2$, $z_2 = -1 + 2i$ and $z_3 = 1 + i$ are plotted in Figure 1, on an Argand diagram for the complex plane with $z = x + iy$

- (a) Explain why z_1 , z_2 and z_3 cannot all be roots of a quartic polynomial equation with real coefficients.

(2)

- (b) Show that $\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \frac{\pi}{4}$

(3)

- (c) Hence show that $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(2)

A copy of Figure 1, labelled Diagram 1, is given on page 12.

- (d) Shade, on Diagram 1, the set of points of the complex plane that satisfy the inequality

$$|z + 2| \leq |z - 1 - i|$$

(2)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

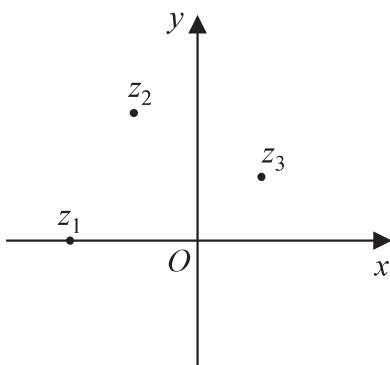


Diagram 1

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 9 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 9 marks)



7. $f(z) = z^3 - 8z^2 + pz - 24$

where p is a real constant.

Given that the equation $f(z) = 0$ has distinct roots

$$\alpha, \beta \text{ and } \left(\alpha + \frac{12}{\alpha} - \beta \right)$$

(a) solve completely the equation $f(z) = 0$ (6)

(b) Hence find the value of p . (2)



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. A gas company maintains a straight pipeline that passes under a mountain.

The pipeline is modelled as a straight line and one side of the mountain is modelled as a plane.

There are accessways from a control centre to two access points on the pipeline.

Modelling the control centre as the origin O , the two access points on the pipeline have coordinates $P(-300, 400, -150)$ and $Q(300, 300, -50)$, where the units are metres.

- (a) Find a vector equation for the line PQ , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, where λ is a scalar parameter.

(2)

The equation of the plane modelling the side of the mountain is $2x + 3y - 5z = 300$

The company wants to create a new accessway from this side of the mountain to the pipeline.

The accessway will consist of a tunnel of shortest possible length between the pipeline and the point $M(100, k, 100)$ on this side of the mountain, where k is a constant.

- (b) Using the model, find

- (i) the coordinates of the point at which this tunnel will meet the pipeline,
(ii) the length of this tunnel.

(7)

It is only practical to construct the new accessway if it will be significantly shorter than both of the existing accessways, OP and OQ .

- (c) Determine whether the company should build the new accessway.

(2)

- (d) Suggest one limitation of the model.

(1)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 12 marks)



9. $f(x) = 2x^{\frac{1}{3}} + x^{-\frac{2}{3}} \quad x > 0$

The finite region bounded by the curve $y = f(x)$, the line $x = \frac{1}{8}$, the x -axis and the line $x = 8$ is rotated through θ radians about the x -axis to form a solid of revolution.

Given that the volume of the solid formed is $\frac{461}{2}$ units cubed, use algebraic integration to find the angle θ through which the region is rotated.

(8)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 8 marks)



10. The population of chimpanzees in a particular country consists of juveniles and adults. Juvenile chimpanzees do not reproduce.

In a study, the numbers of juvenile and adult chimpanzees were estimated at the start of each year. A model for the population satisfies the matrix system

$$\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix} \quad n = 0, 1, 2, \dots$$

where a is a constant, and J_n and A_n are the respective numbers of juvenile and adult chimpanzees n years after the start of the study.

- (a) Interpret the meaning of the constant a in the context of the model.

(1)

At the start of the study, the total number of chimpanzees in the country was estimated to be 64 000

According to the model, after one year the number of juvenile chimpanzees is 15 360 and the number of adult chimpanzees is 43 008

- (b) (i) Find, in terms of a

$$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1}$$

(3)

- (ii) Hence, or otherwise, find the value of a .

(3)

- (iii) Calculate the change in the number of juvenile chimpanzees in the first year of the study, according to this model.

(2)

Given that the number of juvenile chimpanzees is known to be in decline in the country,

- (c) comment on the short-term suitability of this model.

(1)

A study of the population revealed that adult chimpanzees stop reproducing at the age of 40 years.

- (d) Refine the matrix system for the model to reflect this information, giving a reason for your answer.

(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)

(2)



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Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 12 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS

